

Sameer Aryal, Ph.D.

CONTACT INFORMATION

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PROFESSIONAL SUMMARY

Computational and experimental neurobiologist with 10+ years of experience integrating proteomics, transcriptomics, and single-cell data to uncover mechanisms underlying neuropsychiatric and neurodevelopmental disorders. Proven track record of leading cross-functional projects, developing scalable analysis pipelines, and translating multi-omics data into actionable biological insight.

CORE SKILLS

- Multi-omics integration; network biology (WGCNA, graph modeling); meta-analysis
- Synaptic proteomics (DDA, DIA); transcriptomics; single-nucleus RNA-seq (10X, split-pool); translationalomics (Ribo-Seq, Trap-Seq, Ribo-Map)
- Computational analysis (Python, R, MATLAB, Bash); HPC (Slurm/SGE); Linux
- Experimental validation: biochemical purification, molecular assays, imaging

PROFESSIONAL EXPERIENCE

Stanley Center for Psychiatric Research, Broad Institute, Cambridge, MA

Research Scientist

January 2024 – present

- Lead small cross-disciplinary teams conducting computational and experimental studies in genetic mouse models of schizophrenia to identify disease-relevant synaptic and cellular mechanisms.
- Perform meta-analyses of human and mouse synaptic proteomics, integrating transcriptomic and single-nucleus RNA-seq data to infer disrupted molecular pathways.
- Translate multi-omics findings into testable hypotheses and experimentally validate candidate molecular drivers using molecular and biochemical approaches.
- Mentor and supervise research associates, providing scientific direction, code review, and experimental guidance while coordinating efforts across genetics, proteomics, and neurobiology teams.

Postdoctoral Associate, Advisor: Morgan Sheng, M.D., Ph.D. October 2020 – December 2023

- Conducted network-based analyses of human synaptic proteomics (WGCNA), revealing reduced synaptic protein networks and increased autophagy-associated signatures in schizophrenia.
- Developed integrative multi-omics pipelines combining proteomics, transcriptomics, and snRNA-seq in a genetic mouse model of schizophrenia (*Srrm2^{+/-}*).
- Identified altered splicing underlying loss of a key synaptic protein isoform and discovered reduced oligodendrocyte abundance, validated using orthogonal molecular and imaging approaches.

Center for Neural Science, New York University, New York, NY

Graduate Research Assistant

June 2014 – June 2020

- Investigated dysregulated mRNA translation in fragile X syndrome using ribosome profiling and TRAP-Seq.
- Built computational pipelines to quantify ribosome dynamics and protein synthesis rates.

EDUCATION

Ph.D., Basic Medical Science (Neurobiology, Genomics), New York University, 2020 — Advisor: Eric Klann, Ph.D.

Dissertation: “Molecular and computational examination of *de novo* protein synthesis in fragile X syndrome”

B.A., Biology (Honors) & Economics, Williams College, 2012 — Advisor: Tim J. Lebestky,

Ph.D.

Honors Thesis: “The role of DopR neuronal circuits in regulating endogenous arousal in *D. melanogaster*”

SELECTED
PUBLICATIONS

- **Aryal, S.**, Geng, C., Kwon, M. J., Farsi, Z., Goble, N., Asan, A. S., Brenner, K., Shepard, N., Seidel, O., Wang, Y., et al. Srrm2 haploinsufficiency drives SynGAP- γ reduction, Agap3 mis-splicing, and oligodendrocyte deficits in a genetic mouse model of schizophrenia. *Cell Reports*, 45(6), 2026.
- **Aryal, S.**, Bonanno, K., Song, B., Mani, D. R., Keshishian, H., Carr, S. A., Sheng, M., Dejanovic, B. Deep proteomics identifies shared molecular pathway alterations in synapses of schizophrenia and bipolar disorder patients and mouse model. *Cell Reports*, 42(5), 2023.
- Longo, F., **Aryal, S.**, Anastasiades, P. G., et al. Cell-type-specific disruption of cortico-striatal circuitry drives repetitive patterns of behavior in fragile X syndrome model mice. *Cell Reports*, 42(8), 2023.
- Ho, J., Tumkaya, T., **Aryal, S.**, Choi, H., Claridge-Chang, A. Moving beyond P values: data analysis with estimation graphics. *Nature Methods*, 16(7):565–566, 2019.
- **Aryal, S.**, Longo, F., Klann, E. Genetic removal of p70 S6K1 corrects translational dysregulation in fragile X syndrome mice. *PNAS*, 118(18):e2001681118, 2021.
- **Aryal, S.** & Klann, E. Turning up translation in fragile X syndrome. *Science*, 361(6403):648–649, 2018.

Full CV and publication list available at www.sameeraryal.com